lead to a total of +1 rewards, action *b* gets the +1 reward later than action *a*, and since we're discounting the action further in the future, we prefer action *a*. We multiply the +1 reward in action *b* by a weighting factor less than 1, so we lower the reward from +1 to say 0.8, so the choice of action is clear.

The discount factor comes up in real life as well as RL. Suppose someone offers you $100 now or $110 one month from now. Most people would prefer to receive the money now, because we discount the future to some degree, which makes sense because the future is uncertain (what if the person offering you the money dies in two weeks?). Your discount factor in real life would depend on how much money someone would have to offer you in one month for you to be indifferent to choosing that versus getting $100 right now. If you would only accept $200 in a month versus $100 right now, your discount factor would be $100/$200 = 0.5 (per month). This would mean that someone would have to offer you $400 in two months for you to choose that option over getting $100 now, since we'd discount 0.5 for 1 month, and 0.5 again for the next month, which is $0.5 \times 0.5 = 0.25$, and $100 = 0.25x$, so $x = 400$. Perhaps you might see the pattern that discounting is exponential in time. The value of something at time t with a discount factor of γ:[0,1) is γ^t.

The discount factor needs to be between 0 and 1, and we shouldn't set it exactly equal to 1, because if we don't discount at all, we would have to consider the future rewards infinitely far into the future, which is impossible in practice. Even if we discount at 0.99999, there will eventually come a time beyond which we no longer consider any data, since it will be discounted to 0.

In Q-learning, we face the same decision: how much do we consider future observed rewards when learning to predict Q values? Unfortunately, there's no definitive answer to this, or to setting pretty much any of the hyperparameters we have control over. We just have to play around with these knobs and see what works best empirically.

It's worth pointing out that most games are *episodic*, meaning that there are multiple chances to take actions before the game is over, and many games like chess don't naturally assign points to anything other than winning or